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Subject Code: BTCOC602

Subject Name: Computer Networks

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Total Marks: 60 Marks

Study & Revision Guide: Question No. 1 is compulsory (12 marks). Answer any 4 questions from Questions 2 to 6. This document contains comprehensive, highly structured solutions mapped perfectly with system design guidelines.

QUESTION NO. 1 (Compulsory Objective Pack)

[12 Marks]

No.	Question Statement & Options	Correct Answer & Technical Context
1	A computer network is mainly used for: a. Data storage b. Resource sharing c. Program execution d. Hardware manufacturing	b. Resource sharing Networks interconnect independent computational endpoints primarily to share assets, hardware resources (printers, storage), and dynamic datasets.
2	Connection-oriented service is best represented by: a. UDP b. IP c. TCP d. ICMP	c. TCP TCP handles explicit 3-way handshakes to initialize sequences, managing state, data tracking, acknowledgments, and error-recovery structures.
3	TCP/IP model consists of: a. 7 layers b. 5 layers c. 4 layers d. 3 layers	c. 4 layers The standard original internet architecture framework groups networking logic into exactly 4 clear layers: Application, Transport, Internet, and Network Access.
4	Wi-Fi standard belongs to: a. 802.3 b. 802.5 c. 802.11 d. 802.16	c. 802.11

No.	Question Statement & Options	Correct Answer & Technical Context
		IEEE 802.11 provides wireless over-the-air communication specifications covering physical and medium access management configurations.
5	Bluetooth standard is: a. 802.11 b. 802.15.1 c. 802.16 d. 802.3	b. 802.15.1 IEEE 802.15 working groups specialize in Wireless Personal Area Networks (WPAN), where 802.15.1 tracks standard Bluetooth specifications.
6	Frame Relay operates at: a. Physical layer b. Data link layer c. Network layer d. Transport layer	b. Data link layer Frame Relay is an efficient packet-switched WAN technology tracking frame connections and physical link boundaries inside Layer 2.
7	Flow control ensures: a. Error-free transmission b. Sender does not overwhelm receiver c. Faster transmission d. Security	b. Sender does not overwhelm receiver It balances rates by using structural feedback loops (like sliding window credits) to prevent high-performance senders from over-allocating buffers at slower nodes.
8	Routers operate mainly at: a. Data link layer b. Network layer c. Transport layer d. Application layer	b. Network layer Routers process explicit logical addressing frameworks (Layer 3 IP headers) to compute path selections and route packets across multiple networks.
9	UDP is: a. Reliable b. Connection-oriented c. Connectionless d. Secure	c. Connectionless UDP functions completely without sequence state setups, handshakes, or delivery verifications, treating every data block as a fully isolated item.
10	POP protocol is mainly for: a. Sending mail b. Receiving mail c. File upload d. Web browsing	b. Receiving mail The Post Office Protocol (POP) is an application layer mechanism designed specifically to retrieve mail items down from centralized mail servers to local clients.
11	Public key cryptography uses: a. One shared key b. Two keys c. No key d. Password only	b. Two keys Asymmetric algorithms depend on mathematically linked key pairs: a Public key available for anyone

No.	Question Statement & Options	Correct Answer & Technical Context
		to encrypt data, and a Private key kept secure to decrypt it.
12	Firewalls are used to: a. Increase bandwidth b. Control network access c. Detect errors d. Speed transmission	b. Control network access Firewalls act as perimeter control nodes, monitoring and blocking packet traffic based on explicit security rule configurations.

QUESTION NO. 2 (Network Architecture Foundations)

[12 Marks]

A) Explain applications of computer networks in detail.

[6 Marks]

Computer networks form the foundational framework of modern digital interactions across commercial, industrial, and personal applications. Key use cases include:

- **Commercial Resource Sharing:** Interconnects organizational databases, compute clusters, and physical equipment (like storage systems and high-end enterprise printers) across local or distributed setups.
- **Distributed Information Access:** Supports the global web infrastructure, allowing real-time lookups across remote cloud instances, electronic finance nodes, and streaming platforms.
- **Interpersonal Communication:** Powers everyday communications like email, instant messaging systems, Voice over IP (VoIP), and interactive video conferencing tools.
- **E-Commerce and Financial Infrastructure:** Enables secure real-time payments, online shopping ecosystems, electronic document exchanges, and banking clearings.

B) Explain TCP/IP reference model and compare it with OSI model.

[6 Marks]

The **TCP/IP model** is the functional architecture of the internet, structured into 4 foundational layers:

1. **Application Layer:** Interfaces directly with application processes (e.g., HTTP, FTP, DNS, SMTP).
2. **Transport Layer:** Manages end-to-end host transport dynamics via connection-oriented (**TCP**) or connectionless (**UDP**) configurations.
3. **Internet Layer:** Handles packet routing across networks using logical IP addressing rules (**IPv4** , **IPv6**).
4. **Network Access Layer:** Defines the physical radio links, cable connections, and hardware mapping protocols (Ethernet, Wi-Fi MAC layer).

Feature	OSI Reference Model	TCP/IP Functional Model
Total Layers	7 Layers (Application, Presentation, Session, Transport, Network, Data Link, Physical).	4 Layers (Application, Transport, Internet, Network Access).
Development	Designed by ISO as an abstract theoretical standard guide.	Evolved directly from ARPANET research to solve practical hardware needs.
Layer Approach	Strictly decouples data styling (Presentation) and session tracking (Session).	Merges presentation and session handling directly inside the main Application Layer.
Protocols	Loosely defined before building hardware; less adaptable to shifts.	Built natively around established protocols (TCP and IP), making it the global standard.

QUESTION NO. 3 (Local Area Network Technologies)

[12 Marks]

A) Explain Ethernet architecture and working.

[6 Marks]

Ethernet (IEEE 802.3) is the standard protocol framework for wired local area networks.

1. Architecture & Layers: It splits the Data Link Layer into two functional sublayers: the *Logical Link Control (LLC)* for software data framing/flow management, and the *Media Access Control (MAC)* for mapping local hardware destination addresses.

2. Core Working Principle (CSMA/CD): Ethernet networks traditionally balance access using Carrier Sense Multiple Access with Collision Detection:

- **Carrier Sense:** Before transmitting, a node listens to the physical cable to ensure no other station is actively sending data.
- **Multiple Access:** Multiple network devices share the same physical cable layout.
- **Collision Detection:** If two nodes transmit at the same time, a data collision occurs. The stations detect the voltage spike, instantly stop transmitting, send a high-frequency jam signal, and wait a random period using a *Binary Exponential Backoff* algorithm before trying again.

B) Explain wireless LAN technologies: Wi-Fi and Wi-MAX.

[6 Marks]

1. Wi-Fi (IEEE 802.11): A short-range wireless networking technology that connects endpoints within home, office, or campus environments. Because wireless radios cannot transmit and listen on the same channel simultaneously, Wi-Fi uses **CSMA/CA (Collision Avoidance)** instead of collision detection. Devices use precise *Request to Send (RTS)* and *Clear to Send (CTS)* handshakes to reserve the wireless channel before transmitting data.

2. Wi-MAX (IEEE 802.16): Worldwide Interoperability for Microwave Access is a long-range wireless solution designed to deliver high-speed broadband across wide metropolitan areas (WMAN). Unlike Wi-Fi's short-range coverage, Wi-MAX towers provide long-range connectivity over several kilometers, making it an excellent alternative for last-mile internet delivery to rural or underserved regions.

QUESTION NO. 4 (Data Link Layer Principles)

[Solve Any Two • 12 Marks]

A) Explain Data Link Layer design issues in detail.

[6 Marks]

The Data Link Layer (Layer 2) converts a raw physical transmission facility into an organized, reliable communication link. Its main design challenges include:

- Service Interface to Network Layer:** Providing clear, logical service frameworks to Layer 3, including unacknowledged connectionless service, acknowledged connectionless service, or reliable connection-oriented options.
- Framing:** Splitting the raw character streams sent by the physical layer into distinct, identifiable data units called frames.
- Flow Control:** Implementing feedback loops to prevent fast senders from overloading the buffering capacity of slower receivers.
- Error Control:** Detecting and correcting transmission errors caused by line noise or attenuation, typically using frame retransmissions.

B) Explain framing, flow control and error control.

[6 Marks]

- Framing:** Isolates binary bitstreams using specialized strategies like *Character Counts* (storing the exact length in a header field), *Byte Stuffing* (wrapping frames in FLAG delimiters and escaping data flags with ESC prefixes), or *Bit Stuffing* (inserting an extra 0 bit after five consecutive 1 bits to ensure data patterns don't conflict with control frames).
- Flow Control:** Prevents buffer overruns. Common strategies include *Stop-and-Wait* (the sender transmits one frame and pauses until it receives an explicit acknowledgment) and *Sliding Window*

mechanisms (allowing the sender to transmit a specified window of multiple frames before pausing for an acknowledgment).

- **Error Control:** Ensures reliable data delivery by combining error-detection codes with timers and *Automatic Repeat reQuest (ARQ)* protocols. If a frame is corrupted or lost, the receiver rejects it, and the sender retransmits the data block after its acknowledgment timer expires.

C) Compare error detecting codes and error correcting codes.

[6 Marks]

Property	Error Detecting Codes	Error Correcting Codes
Primary Function	Identifies whether a data block was corrupted during transmission, without pinpointing exactly where the flipped bits are.	Not only detects corruption but also determines the precise bit positions that changed, allowing them to be corrected inline.
Data Overhead	Requires minimal extra bits (e.g., adding a single parity bit or a short CRC checksum).	Requires significantly more parity check bits to isolate specific bit locations.
Common Algorithms	Simple Parity checks, Internet Checksums, and cyclic redundancy checks (CRC).	Hamming Codes, Reed-Solomon codes, and Forward Error Correction (FEC) models.
Use Cases	Ideal for reliable, low-noise links (like fiber optics) where retransmitting lost frames is fast and cheap.	Essential for high-latency or extremely noisy connections (like satellite links) where retransmissions are slow and expensive.

QUESTION NO. 5 (Routing & Transport Protocols)

[Solve Any Two • 12 Marks]

A) Explain routing algorithms: distance vector and link state.

[6 Marks]

1. Distance Vector Routing (e.g., RIP): Nodes periodically share their entire local routing tables with their immediate neighbors. Routes are calculated mathematically using the *Bellman-Ford equation*. While simple to implement, distance vector networks suffer from slow convergence and are prone to routing loops, such as the infamous *count-to-infinity* problem.

2. Link State Routing (e.g., OSPF): Nodes broadcast small link-state advertisements (LSAs) containing info about their immediate connections to all routers across the network. Every router compiles these updates to build an identical, complete map of the network topology, then uses *Dijkstra's Shortest Path*

First (SPF) algorithm to calculate its local routing paths. This approach converges rapidly and prevents loops, though it requires more memory and processing power.

B) Explain TCP and UDP with comparison.

[6 Marks]

TCP (Transmission Control Protocol) provides a reliable, connection-oriented byte stream, while **UDP (User Datagram Protocol)** offers a lightweight, connectionless, unacknowledged transport service.

Property	TCP Protocol	UDP Protocol
Connection State	Connection-oriented; requires explicit 3-way handshakes before sending data.	Connectionless; sends isolated data blocks instantly without any handshake state.
Reliability	Guaranteed delivery; automatically manages packet ordering, tracking, and retransmissions.	Best-effort delivery; packets can be lost, delayed, or arrive out of order without notice.
Header Size	Large overhead (20 to 60 bytes) to track sequences and acknowledgments.	Minimal overhead (exactly 8 bytes) for fast processing.
Traffic Control	Includes built-in flow control and congestion avoidance algorithms.	No built-in traffic control; transmits data continuously at the application's source rate.

C) Explain Quality of Service (QoS) and its components.

[6 Marks]

Quality of Service (QoS) refers to a set of network technologies designed to manage traffic patterns and meet strict performance guarantees for critical services like voice and video.

Key QoS Performance Parameters:

- **Reliability:** Minimizing packet loss rates during transit.
- **Delay (Latency):** The time it takes for a packet to travel from its source to its destination.
- **Jitter:** The variance in arrival times between consecutive packets, which can distort real-time audio and video feeds.
- **Bandwidth:** The data transmission capacity allocated to specific traffic classes.

Core Traffic Management Components: Networks enforce QoS using *Traffic Shaping* tools (like the Leaky Bucket algorithm for steady output rates or Token Buckets for managing burst traffic), along with packet prioritization, scheduling, and resource reservation protocols.

QUESTION NO. 6 (Application Layer Services & Security) [Solve Any Two • 12 Marks]

A) Explain DNS and HTTP in detail.

[6 Marks]

1. DNS (Domain Name System): Acts as the internet's directory by translating human-readable domain names (like `www.google.com`) into machine-readable IP addresses (like `142.250.190.46`). It operates as a distributed, hierarchical database split into Root zones, Top-Level Domains (TLD like `.com`, `.org`), and Authoritative servers, resolving queries through recursive or iterative lookups.

2. HTTP (Hypertext Transfer Protocol): The core application layer protocol used to fetch web resources. It follows a simple request-response format over TCP port 80 (or port 443 for secure HTTPS). Clients send standardized request verbs (like GET, POST, PUT), and web servers return structured status codes (such as 200 OK or 404 Not Found) alongside the requested data payload.

B) Explain network security concepts and authentication.

[6 Marks]

Network security ensures the protection of data integrity, confidentiality, and access control across interconnected networks. Its core pillars are summarized by the ****CIA Triad****:

- **Confidentiality:** Preventing unauthorized access to sensitive data, typically achieved through strong encryption algorithms.
- **Integrity:** Ensuring that data cannot be altered or tampered with during transit without detection, using cryptographic hash functions.
- **Availability:** Ensuring that network resources remain accessible to authorized users, protecting against Denial-of-Service (DoS) attacks.

Authentication Mechanisms: Cryptographic authentication verifies identity using mechanisms like digital signatures, multi-factor authorization parameters, and challenge-response protocols. This ensures that communicating nodes can confirm each other's identities before exchanging data, preventing spoofing or man-in-the-middle attacks.

C) Explain digital signatures, certificates and firewalls in detail.

[6 Marks]

Modern network security relies on a combination of host defenses and cryptographic verification systems:

1. **Digital Signatures:** Provide non-repudiation and verify data integrity. The sender runs their message through a hash function and encrypts the resulting digest using their private key. The receiver decrypts this signature using the sender's public key and compares the hashes; a match proves the message was authentic and unaltered.
2. **Digital Certificates (X.509):** Bind a public key to its verified owner. These credentials are issued and cryptographically signed by a trusted third-party *Certificate Authority (CA)*, allowing clients to securely verify a server's identity before initializing an encrypted HTTPS session.
3. **Firewalls:** Act as network perimeter guards by analyzing incoming and outgoing packet headers. They enforce strict access rules based on criteria like source/destination IP addresses, port choices, and network protocols, blocking malicious traffic before it can reach internal systems.